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INTERN MANAGEMENT SYSTEM

Project Documentation

A Complete Web-Based Solution for Intern Task & File Management

Item	Details
Technology	Blazor Server (.NET 10)
Database	SQL Server (SSMS)
ORM	Entity Framework Core
UI Framework	Bootstrap 5
Language	C# 13
Year	2026

1. Problem Statement

Organizations that hire interns often struggle to manage them efficiently. Without a dedicated digital system, all intern-related activities such as task assignment, progress monitoring, and file sharing are carried out manually. This creates significant operational challenges.

- Intern tasks were tracked manually using Excel sheets, paper forms, or verbal communication, leading to confusion and missed deadlines.
- There was no reliable way for supervisors to send reference materials or files to interns, and vice versa.
- Supervisors had no real-time visibility into the status of tasks assigned to interns — whether they were pending, in progress, or completed.
- Admins had no central system to manage user accounts, assign roles, or maintain intern/supervisor profiles.
- There was no separation of access — any user could potentially view or modify data they were not authorized to see.
- Profile management was non-existent — no photos, no department info, no contact details were stored or visible.

These problems resulted in poor communication, delayed task completion, and a lack of accountability within intern programs.

2. Proposed Solution

We developed the Intern Management System (IMS) — a full-stack web application using Blazor Server and SQL Server that addresses all the above problems through a centralized, role-based, and user-friendly platform.

2.1 Role-Based Access Control

- Three distinct roles: Admin, Supervisor, and Intern, each with their own login and dashboard.
- LoginSession (Singleton service) maintains login state across the application.
- Unauthorized users are redirected to the login page automatically.

2.2 Task Management

- Supervisors can assign tasks with title, description, intern selection, due date, and file attachment.
- Tasks are stored in the SQL Server database and displayed on both Supervisor and Intern dashboards.
- Supervisors can delete tasks; interns can view their assigned tasks in real-time.

2.3 File Sharing

- Supervisors can attach files when assigning tasks — files saved to `wwwroot/uploads/` folder.
- File records (name, path, assigned intern, uploader) stored in SupervisorFiles database table.
- Interns see a dedicated file list on their dashboard with a Download button for each file.

2.4 User Management

- Admin can create, edit, and delete both intern and supervisor accounts.
- Edit User page supports updating all profile fields including profile photo upload.
- Users are displayed as visual cards with photo, role badge, and full details.

2.5 Profile Photo System

- Photos uploaded via InputFile component, saved to `wwwroot/uploads/photos/`.
- PhotoPath stored in database and displayed on user cards across all pages.
- Instant preview shown before saving — users can see their photo before confirming.

3. Modules & Functionality Details

3.1 Authentication Module

Feature	Description
Admin Login	Separate login page for admin — redirects to Admin Dashboard on success
Supervisor Login	Supervisor credentials checked — redirects to Supervisor Dashboard
Intern Login	Intern credentials verified — redirects to Intern Dashboard
LoginSession	Singleton service storing Username, FullName, Role across all pages
Route Guard	Each dashboard checks LoginSession.Role — redirects to /login if unauthorized

3.2 Admin Module

Feature	Description
Show All Users	Displays all users as profile cards with photo, role badge, and info
Create Intern	Form to register a new intern with all profile fields + photo upload
Create Supervisor	Form to register a new supervisor account
Edit User	Edit any user's details including photo, phone, department, start date
Delete User	Remove a user from the system permanently
View Interns	Card view of all interns with their details
View Supervisors	Card view of all supervisors with their details

3.3 Supervisor Module

Feature	Description
Supervisor Dashboard	Shows all assigned tasks in a table with delete option
Assign Task	Form to assign task with title, description, intern, due date & file

File Upload	Attach a file when assigning a task — sent directly to the selected intern
Delete Task	Remove a task from the system
View Interns	Browse intern profiles

3.4 Intern Module

Feature	Description
Intern Dashboard	Shows assigned tasks and files received from supervisor
Task List	Table showing task title, description, assigned by, status, due date
File Download	List of files sent by supervisor with a Download button per file

3.5 File Management Module

Feature	Description
File Upload (Task)	Supervisor attaches file when assigning task via InputFile component
File Storage	Files saved to wwwroot/uploads/ with GUID-prefixed unique filenames
DB Record	SupervisorFiles table stores: FileName, FilePath, AssignedTo, UploadedBy, UploadDate
File Download	Intern dashboard shows file list; each row has an anchor tag with download attribute
Photo Upload	User photos saved to wwwroot/uploads/photos/ — path stored in Users.PhotoPath

3.6 Profile Card System

Page	Card Contents
Show Users (/showusers)	Photo, Full Name, Username, Role badge, Gender, Nationality, Department, Supervisor + Edit/Delete buttons
Interns (/interns)	Photo, Full Name, Username, Intern badge, Gender, Nationality, Department, Supervisor, Phone, Start Date
Supervisors (/supervisors)	Photo, Full Name, Username, Supervisor badge, Gender, Nationality, Department, Phone

4. Database Design

Database Name: InternManagementDB (SQL Server)

4.1 Tables Overview

Table	Purpose	Key Columns
Users	All system users	UserId, Username, Password, FullName, Role, Department, PhotoPath
Tasks	Assigned tasks	TaskId, Title, AssignedTo, AssignedBy, DueDate, Status
SupervisorFiles	Files sent by supervisors	Id, FileName, FilePath, AssignedTo, UploadedBy
InternFiles	Files uploaded by interns	Id, FileName, FilePath, InternName, SupervisorName
Attendance	Daily attendance	Id, InternName, Date, Status
ActivityLog	System activity log	Id, Message, CreatedAt

4.2 Technology Stack

Technology	Version / Details	Purpose
Blazor Server	.NET 10	Full-stack web framework — C# for both frontend and backend
SQL Server	SSMS 2022	Relational database for all application data
Entity Framework Core	Latest	ORM — maps C# models to database tables
IDbContextFactory	.NET 10	Thread-safe database access for Blazor Server
Bootstrap 5	CDN	Responsive UI components — cards, tables, forms, buttons
Chart.js	CDN	JavaScript charts rendered via Blazor IJSRuntime interop
InputFile	Blazor Built-in	File upload component for photos and task attachments

5. System Interfaces

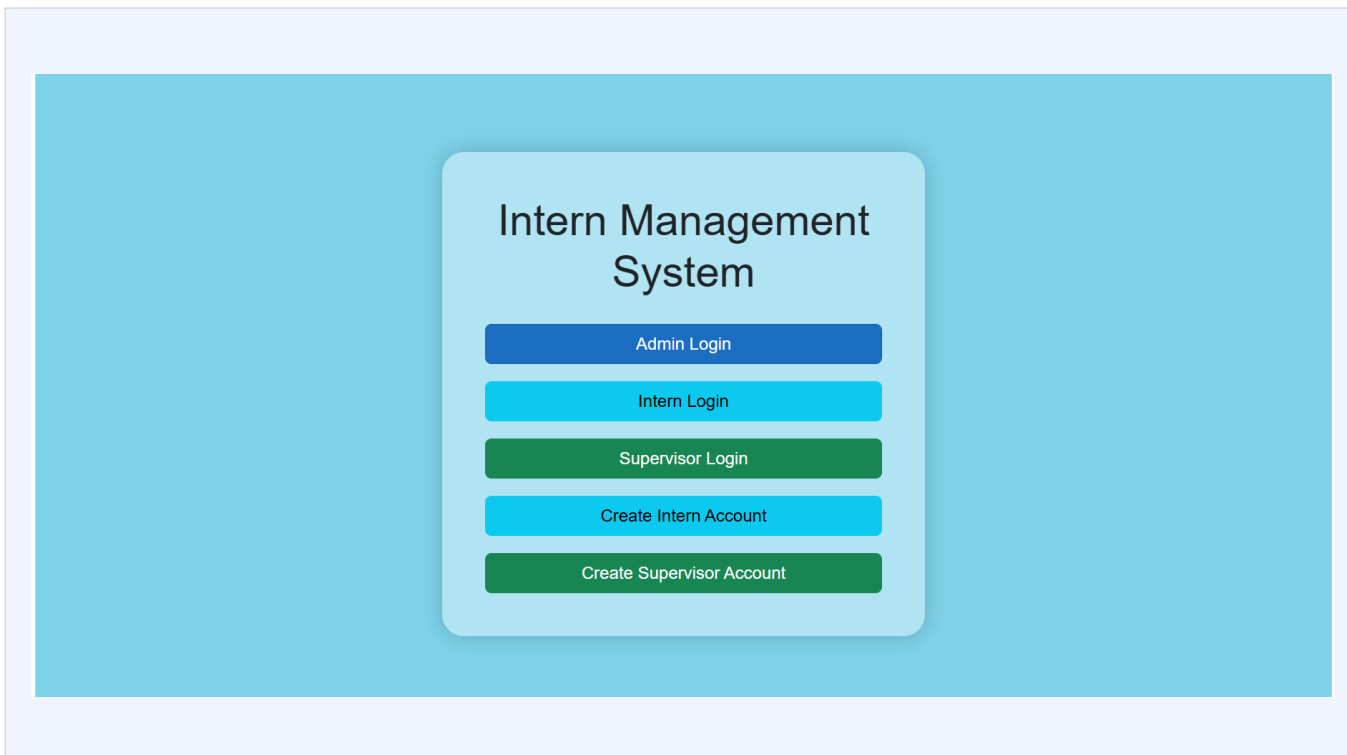
This section presents each interface of the Intern Management System along with a detailed explanation of its purpose, features, and how it works. Each interface is designed with a specific role in mind and contributes to the overall functionality of the system.

5.1 Login

The Login Portal is the entry point of the Intern Management System. When a user first opens the application, they are greeted by this screen. It presents three distinct login options: Admin Login, Supervisor Login, and Intern Login. Each button redirects the user to a role-specific authentication page.

This design ensures clean separation of roles from the very first interaction. Instead of a single login form that determines the role after entry, the system routes each type of user separately, making the experience clearer and reducing the risk of wrong-role access. Additionally, quick-access buttons for creating new Intern and Supervisor accounts are available on this screen for onboarding purposes.

- Provides three separate login paths for Admin, Supervisor, and Intern.
- Includes buttons to create new Intern and Supervisor accounts directly.
- Clean and minimal design ensures fast navigation without confusion.

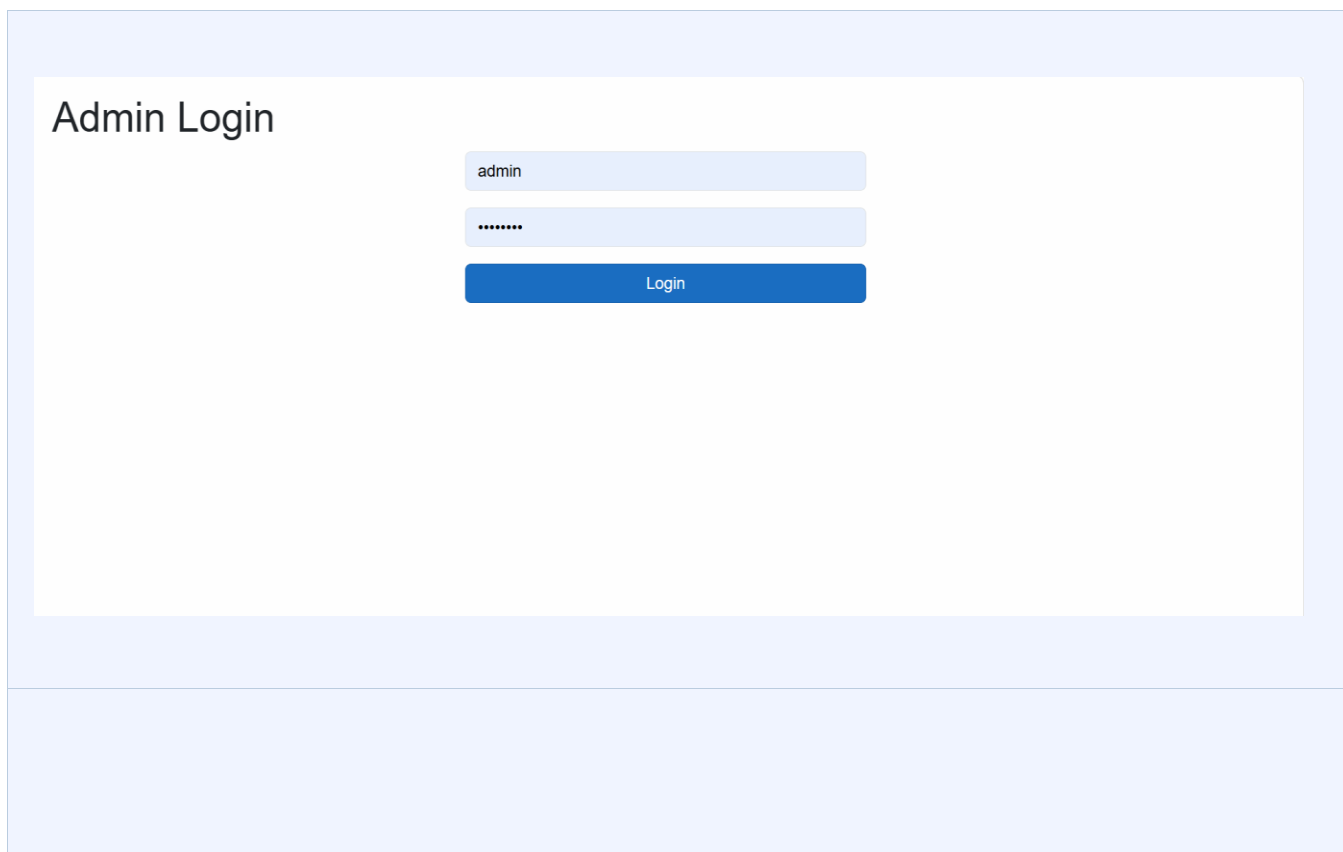


5.2 Admin Login

The Admin Login page provides a secure authentication form exclusively for administrators. The admin enters their username and password, which are validated against the Users table in the database where Role = 'Admin'. Upon successful authentication, the LoginSession service stores the admin's credentials and the system redirects to the Admin Dashboard.

This dedicated login page prevents interns and supervisors from accidentally or intentionally accessing the admin panel. The LoginSession singleton ensures that the admin's identity is preserved throughout the session without the overhead of traditional cookie-based authentication.

- Validates admin credentials against the SQL Server Users table.
- Sets LoginSession.Role = 'Admin' on successful login.
- Redirects to /dashboard on success, shows error on failure.

A screenshot of the Admin Login page. The page has a light blue background. In the center, there is a white rectangular box containing the login form. The form is titled "Admin Login" in a large, dark font. Below the title, there are two input fields: the first is for the username, containing the text "admin", and the second is for the password, containing seven dots. Below these fields is a blue button with the text "Login" in white.

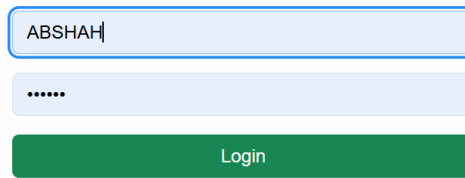
5.3 Supervisor Login

The Supervisor Login page authenticates supervisors into the system. Supervisors are staff members responsible for assigning tasks, uploading files, and monitoring intern progress. After entering valid credentials, the LoginSession is updated with the supervisor's full name and role, and the user is redirected to the Supervisor Dashboard.

The supervisor's FullName stored in LoginSession is critical — it is automatically used as the 'AssignedBy' field whenever a task is created, and as the 'UploadedBy' field when a file is sent to an intern. This removes the need for supervisors to manually type their name on every form.

- Authenticates supervisor credentials against the database.
- LoginSession.FullName is used automatically in task and file operations.
- Redirects to /supervisor-dashboard on successful login.

Supervisor Login

A screenshot of a web form titled "Supervisor Login". It features two input fields: the first contains the text "ABSHAH" and the second contains six dots, indicating a password field. Below these fields is a green button with the text "Login".

5.4 Intern Login

The Intern Login page allows registered interns to access their personal dashboard. After authentication, the intern's full name is stored in LoginSession and used to filter tasks and files that belong specifically to that intern. This ensures that each intern only sees their own assigned tasks and received files — not those of other interns.

This is a fundamental privacy and security feature of the system. Since task filtering and file retrieval are based on LoginSession.FullName, it is critical that intern names match exactly between the Users table and the Tasks/SupervisorFiles tables.

- Filters tasks and files using LoginSession.FullName after login.
- Each intern sees only their own data — no cross-intern data leakage.
- Redirects to /intern-dashboard on successful authentication.

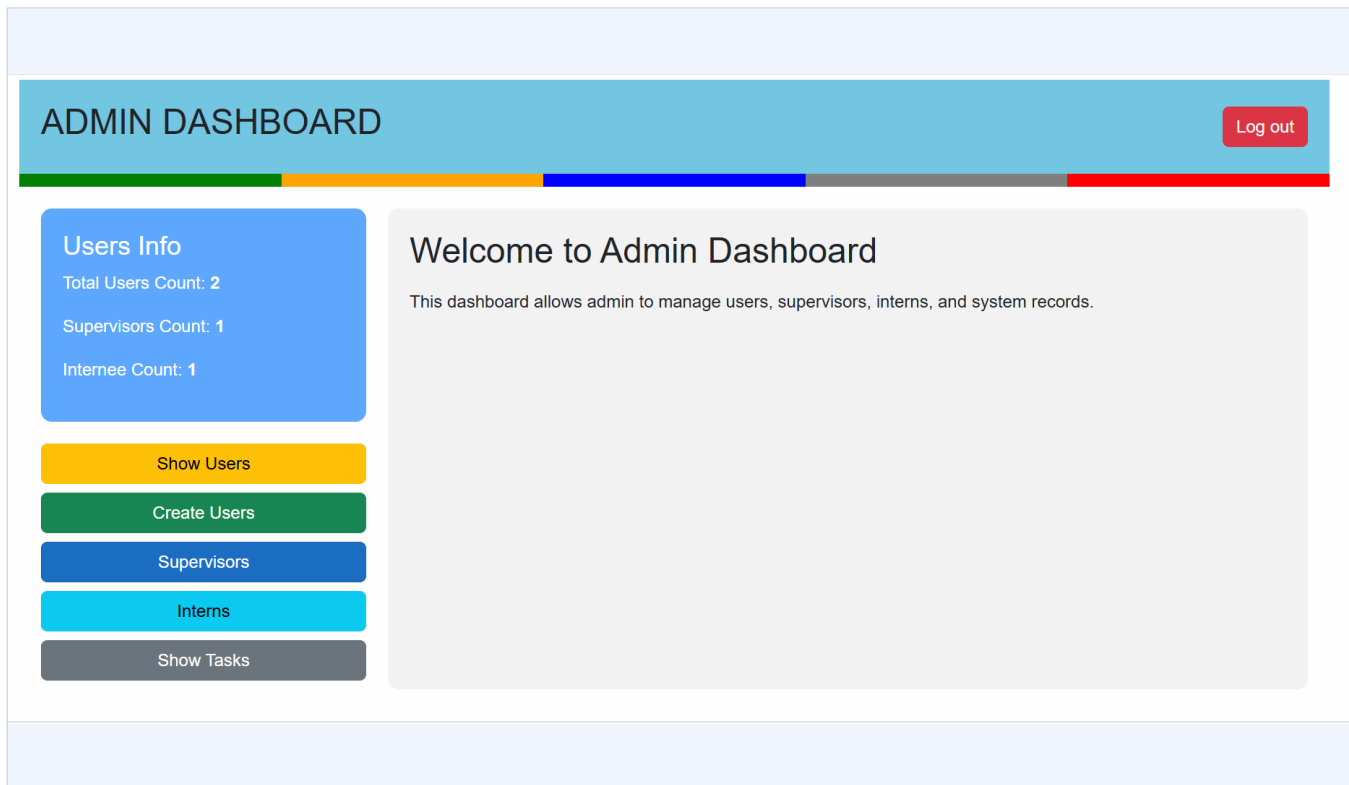
Intern Login

5.5 Admin Dashboard

The Admin Dashboard is the central control panel for system administrators. It provides quick navigation links to all administrative functions including user management, intern listings, supervisor listings, and system-wide task views. The admin has the highest level of access in the system and can perform all CRUD (Create, Read, Update, Delete) operations on users.

The dashboard is designed to give the admin a bird's-eye view of the system. From here, the admin can quickly jump to any section — whether it is creating a new intern account, editing an existing user's profile, or reviewing the full list of supervisors. The clean navigation layout ensures that administrative tasks can be completed efficiently.

- Central hub for all administrative operations in the system.
- Provides navigation to: Show Users, Create Intern, Create Supervisor, View Interns, View Supervisors.
- Protected page — only accessible when `LoginSession.Role == 'Admin'`.



5.6 Supervisor Dashboard

The Supervisor Dashboard is the primary workspace for supervisors. It displays a table of all tasks that have been assigned in the system, showing the task title, description, the intern it was assigned to, and the current status. The supervisor can delete any task directly from this table using the Delete button.

At the top of the dashboard, an 'Assign Task' button provides quick access to the task assignment form. This workflow — view all tasks, identify pending ones, and assign new ones — is the core operational loop for supervisors. The dashboard is kept intentionally simple so that supervisors can focus on task management without distraction.

- Displays all assigned tasks in a structured table with status badges.
- Delete button on each row allows immediate task removal.
- 'Assign Task' button links to the task assignment form.
- Protected — only accessible when `LoginSession.Role == 'Supervisor'`.

Supervisor Dashboard

Assign Task

Assigned Tasks

Task Title	Description	Assigned To	Status	Action
DBMS	DBMSSK	M.ABDULLAH	Pending	Delete

5.7 Assign Task Page

The Assign Task page is one of the most feature-rich interfaces in the system. It provides a comprehensive form through which supervisors can assign a task to a specific intern. The form includes fields for Task Title, Task Description, Intern Selection (populated from the database), Supervisor Name, Due Date, and an optional File Attachment.

- Task Title, Description, Intern dropdown, Supervisor Name, Due Date — all in one form.
- File attachment is optional — task can be assigned without a file.
- File saved to `wwwroot/uploads/` with a unique GUID prefix to prevent name conflicts.
- Both Task and File records saved to database in a single submission.
- Redirects to Supervisor Dashboard after successful task assignment.

Assign Task

Task Title

Task Description

Select Intern

Assigned By (Supervisor Name)

06/01/2026

Attach File (optional)

Choose File

No file chosen

Assign Task

5.8 Intern Dashboard

The Intern Dashboard is the personal workspace for each intern. It is divided into two main sections. The first section displays a table of all tasks assigned to the logged-in intern, showing the task title, description, the supervisor who assigned it, the current status, and the due date. The second section shows a list of files that have been sent by the supervisor.

Both the task list and the file list are filtered using `LoginSession.FullName`, which means each intern only sees data relevant to them. The file download section includes a green Download button for each file. When clicked, the browser downloads the file directly from the server's `wwwroot/uploads/` directory. This seamless file delivery mechanism eliminates the need for email attachments or external file sharing services.

- Section 1: Task list filtered by intern name — shows title, description, assigned by, status, due date.
- Section 2: File list with Download button — files sent directly by the supervisor.
- All data is personalized — only the logged-in intern's tasks and files are shown.
- Status badges visually highlight Completed (green) vs Pending (yellow) tasks.

Intern Dashboard

My Assigned Tasks

Task Title	Description	Assigned By	Status	Due Date
DBMS	DBMSSK	ABSHAH	Pending	6/1/2026

Files Sent by Supervisor

File Name	Sent By	Date	Download
PAKISTAN STUDIES 04.pdf		6/1/2026	Download
PAKISTAN STUDIES 04 (1).pdf	ABSHAH	6/1/2026	Download

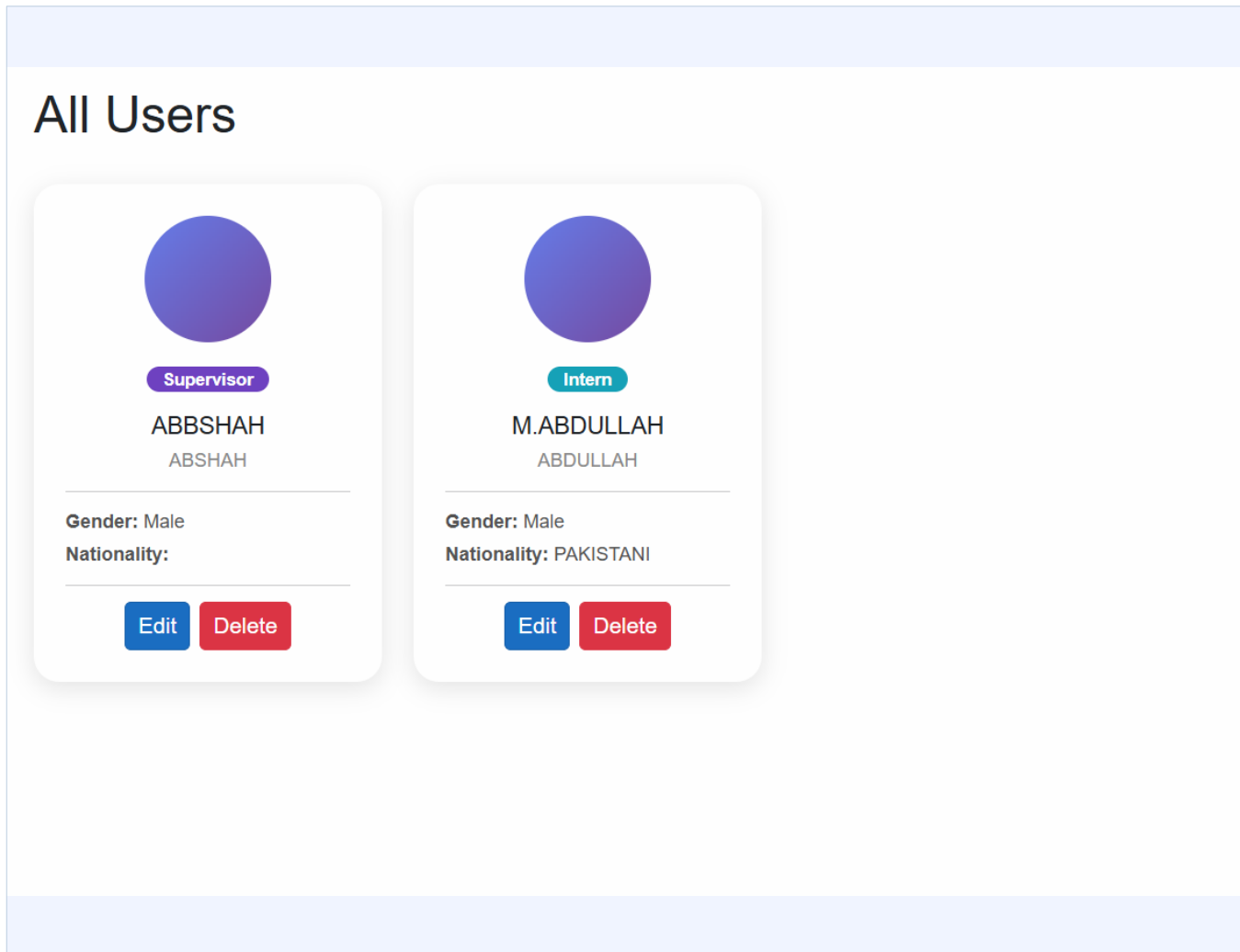
5.9 Show All Users

The Show Users page replaces the traditional data table with a modern card-based layout. Each user — whether Admin, Supervisor, or Intern — is displayed as a profile card. The card shows the user's profile photo (or a default avatar if no photo is set), their full name, username, role badge with color coding, gender, nationality, department, and supervisor name.

Each card also includes two action buttons: an Edit button that navigates to the Edit User page for that specific user, and a Delete button that removes the user from the database immediately. The card layout makes it

significantly easier to scan and identify users visually compared to a plain table, especially in organizations with many interns and supervisors.

- Visual card layout — much more readable than a plain data table.
- Role badges are color-coded: Purple for Admin, Blue for Supervisor, Teal for Intern.
- Profile photo displayed if uploaded; default avatar emoji shown otherwise.
- Edit and Delete buttons on each card for direct user management.



5.10 Interns Page

The Interns page displays only users with the role of 'Intern' in a card-based format. This page is accessible to both Admin and Supervisor roles, allowing supervisors to browse intern profiles when deciding who to assign tasks to. Each card shows the intern's photo, full name, username, department, supervisor name, phone number, and start date.

The additional fields — phone, start date — make this page useful as a quick reference directory for the organization. Supervisors can use it to identify which interns belong to their team and what department they are in, before heading to the Assign Task page.

- Shows only Intern-role users — filtered from the Users table by Role = 'Intern'.
- Displays phone number and start date in addition to standard profile fields.
- Accessible by both Admin and Supervisor for intern browsing.

Interns



Intern

M.ABDULLAH

ABDULLAH

Gender: Male

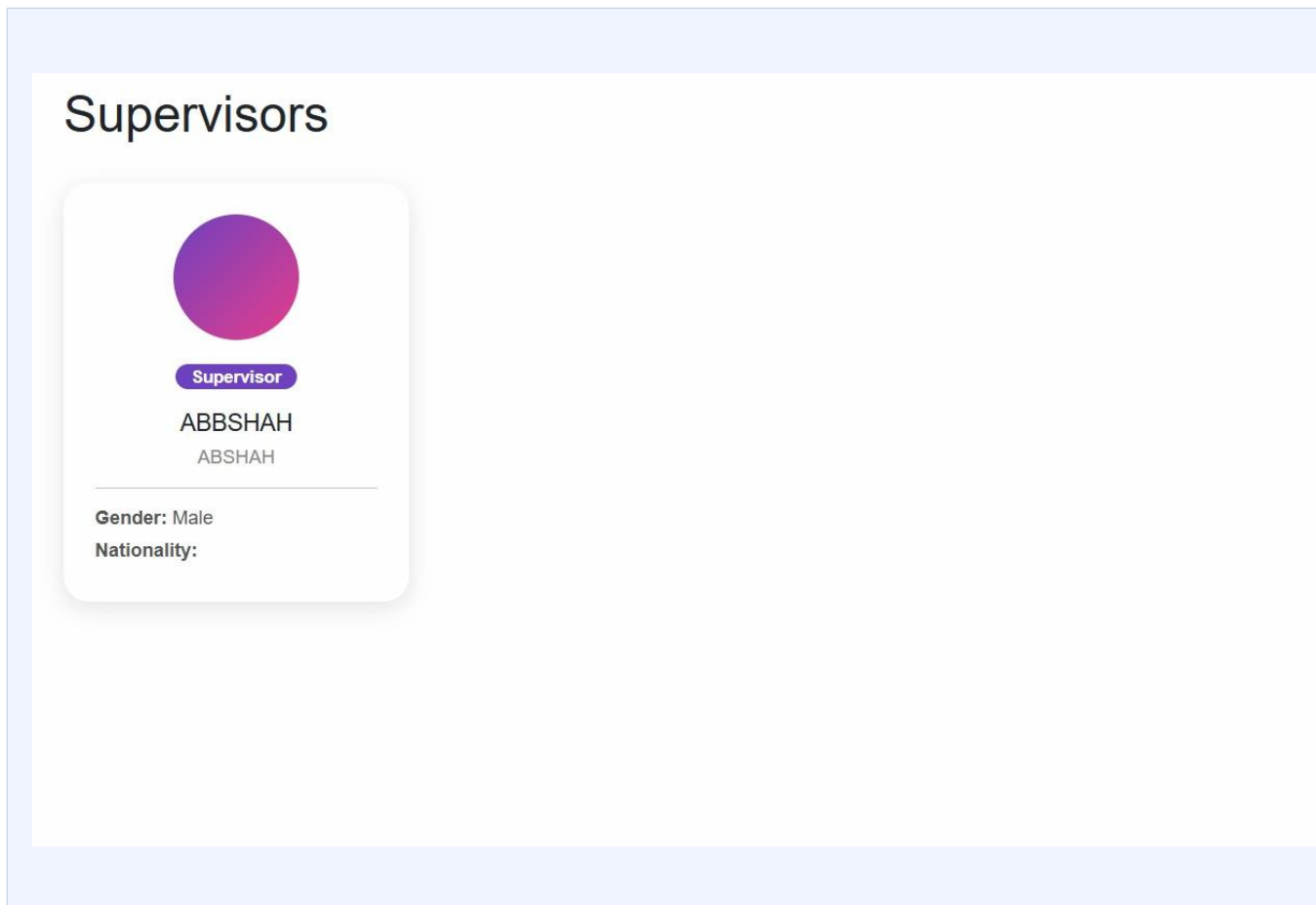
Nationality: PAKISTANI

5.11 Supervisors Page

The Supervisors page displays all users with the role of 'Supervisor' in the same card-based layout. This page is primarily used by the Admin to manage and review supervisor accounts. Each card shows the supervisor's profile photo, full name, username, gender, nationality, department, and phone number.

Having a dedicated supervisors page makes it easy for the admin to identify which supervisors are registered in the system, verify their department assignments, and navigate quickly to edit their profiles if needed. The card layout ensures that supervisor information is presented clearly and consistently.

- Displays only Supervisor-role users — filtered by Role = 'Supervisor'.
- Shows department and phone number for organizational reference.
- Used by Admin to review and manage supervisor accounts.



5.12 Create Intern

The Create Intern page provides a form for administrators to register new intern accounts in the system. The form includes all necessary fields: Username, Password, Full Name, Gender, Nationality, Department, Phone Number, Start Date, Supervisor assignment, and Profile Photo. The supervisor dropdown is dynamically populated from the database, showing only users with the Supervisor role.

When a profile photo is selected, an instant preview is shown above the form so the admin can verify the image before saving. Upon form submission, the photo is saved to `wwwroot/uploads/photos/` and the user record is inserted into the database with all fields including the `PhotoPath`. The new intern can then immediately log in using the credentials set during account creation.

- All profile fields in one form — no need for a separate edit step after creation.
- Supervisor dropdown auto-populated from database — only shows active supervisors.
- Instant photo preview before saving — prevents wrong photo uploads.
- New intern can login immediately after account creation.

Create User

Username

Password

Full Name

Select Gender

Select Role

Nationality

Select Supervisor

Create User

5.13 Edit User

The Edit User page allows administrators to modify any existing user's information. The page loads the current user data from the database based on the `UserId` passed in the URL and pre-fills all form fields. The admin can update any combination of fields including Username, Password, Full Name, Gender, Role, Nationality, Department, Phone, Start Date, Supervisor Name, and Profile Photo.

- Pre-fills all form fields with existing user data from the database.
- Photo update: shows current photo, allows new upload with instant preview.
- Validation ensures required fields (Username, Full Name) are not left empty.
- Updates all fields including Phone, StartDate, PhotoPath in a single save operation.
- Redirects to `/showusers` after successful update.

Edit User

Editing User ID: 1012



6. Database Design

The system uses Microsoft SQL Server as its relational database. The database is named InternManagementDB and contains six tables that together support all features of the application.

6.1 Tables Overview

Table	Purpose	Key Columns
Users	All system users	UserId, Username, Password, FullName, Role, Department, PhotoPath
Tasks	Assigned tasks	TaskId, Title, AssignedTo, AssignedBy, DueDate, Status
SupervisorFiles	Files sent by supervisors	Id, FileName, FilePath, AssignedTo, UploadedBy
InternFiles	Files uploaded by interns	Id, FileName, FilePath, InternName, SupervisorName
Attendance	Daily attendance	Id, InternName, Date, Status
ActivityLog	System activity log	Id, Message, CreatedAt

6.2 Technology Stack

Technology	Version	Purpose
Blazor Server	.NET 10	Full-stack web framework — C# for both frontend and backend
SQL Server	SSMS 2022	Relational database for all application data
Entity Framework Core	Latest	ORM — maps C# models to database tables
DbContextFactory	.NET 10	Thread-safe database access for Blazor Server
Bootstrap 5	CDN	Responsive UI components — cards, tables, forms, buttons
Chart.js	CDN	JavaScript charts rendered via Blazor IJSRuntime interop
InputFile	Blazor Built-in	File upload component for photos and task attachments

7. Project Summary

Item	Detail
Project Name	Intern Management System (IMS)
Type	Web Application — Blazor Server (.NET 10)
Database	SQL Server — InternManagementDB
Total Pages	13+ Razor pages
Total Tables	6 SQL tables
Total Services	6 C# service classes
Roles	Admin, Supervisor, Intern
File Upload	Task attachments + Profile photos
Authentication	Role-based via LoginSession (Singleton)
Status	Functional — runs on localhost:7133

The Intern Management System successfully digitizes and centralizes all intern management operations, providing a clean, role-based interface for admins, supervisors, and interns to collaborate effectively. Every feature from task assignment and file sharing to profile management and visual cards has been implemented with usability and data integrity as the primary goals.